Research Gap Analysis: Artificial Intelligence, Machine Learning, Federated Learning, Large Language Models, and Explainable AI in Cancer and Rare-Disease Diagnosis

1. Scope and Overview of the Reviewed Literature

This research-gap analysis synthesizes five uploaded papers addressing complementary aspects of artificial intelligence (AI) and machine learning (ML) in oncology and rare-disease diagnosis:

1. **Kourou et al. (2015)** — *Machine learning applications in cancer prognosis and prediction*
2. **Pati et al. (2022)** — *Federated Learning Enables Big Data for Rare Cancer Boundary Detection*
3. **Zhang, Shi, and Wang (2023)** — *Machine Learning and AI in Cancer Prognosis, Prediction, and Treatment Selection: A Critical Approach*
4. **Sangwan (2024)** — *Quantifying Explainable AI Methods in Medical Diagnosis: A Study in Skin Cancer*
5. **Shyr et al. (2026)** — *An artificial intelligence framework for end-to-end rare disease phenotyping from clinical notes using large language models*

Collectively, the literature demonstrates a clear progression from conventional predictive ML toward distributed learning, multimodal clinical AI, explainability, and end-to-end workflow-oriented systems. However, this progression has not yet produced a unified framework that simultaneously addresses **generalizability, heterogeneous multimodal data, privacy-preserving learning, explainability, clinical workflow integration, prospective validation, and applicability across both common and rare diseases**.

The central research gap is therefore not simply the absence of more accurate predictive models. Rather, the major unresolved problem is the development and prospective validation of **clinically deployable, generalizable, interpretable, privacy-preserving, multimodal AI systems that can translate heterogeneous patient data into actionable decision support across diverse cancer and rare-disease contexts**.

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2. Literature Synthesis

2.1 Thematic Cluster I: Conventional ML for Cancer Prediction and Prognosis

Kourou et al. (2015)

Kourou et al. provide an early systematic overview of ML applications in cancer susceptibility, recurrence, and survival prediction. The review examines methods including artificial neural networks, Bayesian networks, support vector machines, decision trees, and semi-supervised learning, with inputs spanning

clinical, genomic, histological, demographic, and imaging data. The authors identify a strong trend toward integrating heterogeneous data sources.

The review concludes that ML can improve prediction of cancer susceptibility, recurrence, and survival, but repeatedly emphasizes insufficient external validation, small datasets, unstable models, heterogeneous data challenges, and limited translation into clinical practice. The review specifically notes that small sample sizes relative to feature dimensionality can lead to biased or unstable models and impair generalizability.

Kourou et al. also argue that appropriate statistical treatment of heterogeneous datasets and the construction of broader public cancer databases are necessary for more valid predictive modeling and integrated clinical decision-making.

Zhang, Shi, and Wang (2023)

Zhang et al. extend this perspective by reviewing contemporary applications of AI and ML in cancer diagnosis, prognosis, risk assessment, and treatment selection. Their review covers a wide range of cancers and emphasizes the potential of data-driven methods to improve diagnostic accuracy, prognosis estimation, clinical workflow, and treatment decisions.

At the same time, the authors characterize clinical AI as remaining relatively immature, noting that only a limited number of AI solutions had achieved real-world authorization or adoption at the time of publication. The review also identifies difficulties arising from inaccessible medical data, heterogeneous data sources, normalization requirements, and the interpretability problem associated with complex AI systems.

Synthesis of Cluster I

The two reviews establish the fundamental capability of ML for cancer prediction but reveal a persistent implementation paradox: **predictive capability has advanced faster than clinical validity and deployment readiness**. The literature demonstrates that increasingly sophisticated algorithms do not automatically solve issues of sample adequacy, dataset shift, external validation, data interoperability, interpretability, or clinical integration.

2.2 Thematic Cluster II: Federated Learning and Cross-Institutional Generalizability

Pati et al. (2022)

Pati et al. address one of the limitations identified in earlier cancer-ML research: insufficiently diverse training data. Their study uses federated learning (FL) to train a glioblastoma tumor-boundary segmentation model across **71 geographically distinct healthcare institutions spanning six continents**, involving **6,314 patients and 25,256 MRI scans**. The system uses a 3D-ResUNet architecture and enables participating institutions to retain their data locally while sharing model updates rather than patient-level data.

The study incorporated both local validation and genuinely unseen out-of-sample sites. The final consensus model produced substantial improvements over the initial model, including improvements of 27%, 33%, and 16% for different tumor sub-compartments on local validation data and 15%, 27%, and 16% on unseen data.

This study is particularly important because it provides empirical evidence that geographically distributed and heterogeneous data can improve generalizability while avoiding centralized transfer of sensitive medical data. Its 590-case out-of-sample cohort was deliberately excluded from training, creating a stronger test of generalization than a conventional random train/test partition.

Synthesis of Cluster II

Pati et al. demonstrate that federated learning can mitigate an important weakness of conventional ML: restricted access to heterogeneous multi-institutional datasets. However, the study remains focused on a **specific imaging task in glioblastoma**, rather than an end-to-end clinical decision-support framework. Consequently, FL successfully addresses the *data-distribution and privacy dimension* of the research problem, but not the complete spectrum of clinical AI requirements.

2.3 Thematic Cluster III: Explainable AI and Trustworthy Medical AI

Sangwan (2024)

Sangwan investigates explainability specifically within skin-cancer diagnosis. The study uses the HAM10000 dataset containing 10,015 dermoscopic images from seven lesion categories and compares multiple deep-learning architectures, including custom CNNs, DenseNet, MobileNet, and ResNet. SHAP, LIME, and Integrated Gradients are then applied to generate post-hoc explanations.

The principal contribution is methodological: rather than relying only on visual inspection of attribution maps, the study quantitatively evaluates explanations using measures associated with **faithfulness, robustness, complexity, localization, monotonicity, stability, and input-reduction fidelity**.

The study nevertheless identifies an important conceptual limitation. Pixel-level explanations may not correspond to the semantic features clinicians actually use when diagnosing skin lesions. The authors therefore recommend moving toward segmentation-based or concept-based explanations and incorporating expert user studies and domain-specific diagnostic criteria. They also explicitly identify generalizability to diverse datasets and continuous adaptation as important future directions.

Synthesis of Cluster III

This work shifts the research question from **“Is the model accurate?”** to **“Can we quantitatively evaluate why the model makes its prediction, and is that explanation clinically meaningful?”** However, explainability remains largely evaluated as a computational property rather than as a demonstrated contributor to improved clinician decisions or patient outcomes.

2.4 Thematic Cluster IV: End-to-End LLM-Based Clinical Phenotyping

Shyr et al. (2026)

Shyr et al. represent a newer generation of clinical AI research through RARE-PHENIX, an end-to-end framework for rare-disease phenotyping from unstructured clinical notes. Rather than treating phenotype extraction, ontology normalization, and diagnostic prioritization as isolated tasks, the framework integrates three sequential stages: LLM-based phenotype extraction, ontology-grounded standardization to Human Phenotype Ontology (HPO) terms through retrieval-augmented generation, and supervised ranking of diagnostically informative phenotypes.

The study trained the framework using 2,671 patients from 11 Undiagnosed Diseases Network sites and externally evaluated it using 16,357 real-world clinical notes from Vanderbilt University Medical Center. The framework outperformed PhenoBERT, with an ontology-based similarity of 0.70 versus 0.58 in the reported end-to-end evaluation.

Importantly, the paper identifies several limitations even after achieving external validation. The UDN cohort is enriched for complex multisystem cases, large 70-billion-parameter models may be difficult to deploy in resource-constrained environments, clinician-curated HPO terms may be incomplete or variable between clinicians, and retrospective model performance does not establish improvement in diagnostic accuracy, time-to-diagnosis, or clinician workload.

The systematic error analysis further shows that many errors originate not from simple extraction failure but from linguistic variability, indirect descriptions in clinical notes, undocumented physical findings, and ontology-granularity differences.

Synthesis of Cluster IV

RARE-PHENIX demonstrates a major conceptual advance: **clinical AI should model the workflow that clinicians actually perform rather than optimizing a single computational subtask**. Nevertheless, the work remains concentrated on rare-disease phenotyping from clinical text and does not yet unify multimodal cancer imaging, clinical variables, treatment history, molecular information, federated learning, and clinically validated explanations within one architecture.

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3. Cross-Paper Synthesis

Across the five papers, six broad developments can be identified.

First, predictive performance is no longer the only major research objective. Earlier cancer-ML literature concentrated primarily on prediction accuracy, whereas newer research increasingly considers generalizability, privacy, explainability, and workflow alignment.

Second, **data heterogeneity is both an asset and a major methodological problem**. Kourou et al. identify the value of combining clinical, genomic, imaging, and histological information but also highlight validation and dataset limitations. Pati et al. show that geographically distributed data can improve performance, while Zhang et al. describe data normalization and interoperability as continuing barriers.

Third, **generalizability remains a central unresolved problem**, although the meaning of generalizability has evolved. Earlier studies frequently lacked external validation; Pati et al. address cross-site generalization explicitly through federated and out-of-sample validation, while Shyr et al. demonstrate external validation across institutions. Yet both remain disease- or modality-specific.

Fourth, **explainability has become increasingly quantitative but remains insufficiently clinical**. Sangwan demonstrates that explanation methods can be objectively evaluated, but also notes that computationally faithful explanations may not necessarily correspond to clinically meaningful concepts.

Fifth, **workflow integration is emerging but fragmented**. RARE-PHENIX is explicitly designed around the clinical phenotyping workflow, whereas much cancer AI research still concentrates on individual prediction or classification tasks.

Sixth, there remains a substantial distinction between **retrospective technical validity and demonstrated clinical utility**. This gap is explicit in both the rare-disease and cancer literatures. Shyr et al. state that prospective evaluation is required to establish effects on diagnostic accuracy, time to diagnosis, and clinician workload. Zhang et al. similarly note that only a limited number of AI applications had reached real-world use.

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4. Research Gap Identification

4.1 Methodological Gaps

4.1.1 Insufficient integration of validation paradigms

Older cancer-ML studies commonly rely on internal cross-validation or relatively small datasets, creating a risk that apparent performance reflects dataset-specific characteristics rather than genuine clinical generalization. Kourou et al. explicitly identify lack of external validation, small samples, and inadequate validation as recurring problems.

More recent work demonstrates stronger validation, but the methods remain fragmented. Pati et al. employ geographically unseen sites, while Shyr et al. use a separate institutional external cohort. A systematic framework combining **patient-level, temporal, institution-level, geographic, demographic, and prospective validation** remains underdeveloped.

4.1.2 Limited multimodal methodological integration

The reviewed research considers different modalities—MRI, dermoscopic images, clinical notes, clinical variables, genomic information—but typically does not integrate these modalities into one validated architecture.

This creates a significant methodological gap because cancer diagnosis and treatment decisions are inherently multimodal. A model that processes only imaging, only clinical text, or only structured clinical variables may miss clinically important relationships that emerge across modalities.

4.1.3 Dataset and annotation limitations

The literature demonstrates that ground truth is not necessarily objective. RARE-PHENIX depends on clinician-curated HPO terms that may be incomplete or subject to inter-clinician variability. Sangwan's work also identifies data diversity as a limitation.

A methodological need therefore exists for **multi-expert annotation protocols, uncertainty-aware labels, adjudication procedures, and explicit modeling of annotation disagreement**.

4.1.4 Computational constraints are insufficiently incorporated into model design

High-performing systems frequently use computationally expensive architectures. Shyr et al. report that the strongest RARE-PHENIX performance was achieved using large 70B-parameter models, which may be impractical in resource-constrained environments.

The literature therefore lacks sufficient comparative work on the trade-off between **accuracy, latency, memory consumption, energy requirements, privacy, and clinical usefulness**.

4.2 Theoretical Gaps

4.2.1 Absence of a unified theory of trustworthy clinical AI

The papers separately address prediction, federated learning, explainability, or workflow alignment. What is missing is an integrated theoretical framework that defines clinical AI quality as a multidimensional construct involving:

predictive accuracy + generalizability + uncertainty + explainability + privacy + workflow compatibility + clinical utility.

Current studies tend to optimize individual dimensions rather than model the interactions among them.

4.2.2 Limited theoretical integration between AI reasoning and clinical reasoning

RARE-PHENIX moves toward clinically aligned workflow modeling, while Sangwan demonstrates that mathematical explanation quality does not necessarily imply clinical meaningfulness. These findings indicate a deeper theoretical question: **What constitutes a clinically valid explanation?**

A future theoretical framework should distinguish between:

- computational explanation,
- causal or mechanistic explanation,
- clinically interpretable explanation,
- actionable explanation.

These concepts are often conflated in medical AI.

4.2.3 Insufficient theory for adaptive, longitudinal clinical AI

Most reviewed systems treat prediction as relatively static. However, cancer patients generate longitudinal data through repeated imaging, laboratory tests, clinical notes, pathology, treatment response, and disease progression.

The literature provides insufficient theoretical development regarding **continual learning, temporal adaptation, concept drift, model updating, and longitudinal uncertainty** in real clinical environments.

4.3 Empirical and Contextual Gaps

4.3.1 Underrepresentation of rare cancers and uncommon disease subtypes

Although Pati et al. specifically target glioblastoma and Shyr et al. address rare diseases, most of the wider cancer-AI literature remains concentrated in more data-rich disease areas.

Kourou et al. already identified limitations caused by restricted disease-specific datasets and noted that some predictive models could not generalize across cancer types.

A major empirical opportunity therefore exists in developing models capable of learning across **multiple cancer types while preserving disease-specific distinctions and rare-subtype sensitivity**.

4.3.2 Geographic and demographic representativeness

Pati et al. substantially improve geographic diversity through 71 institutions across six continents. However, geographic participation alone does not guarantee demographic equity.

The reviewed literature provides insufficient evidence concerning performance variation across socioeconomic groups, ethnic populations, age groups, healthcare systems, resource levels, and demographic subgroups.

4.3.3 Limited prospective clinical evidence

This is one of the strongest gaps across all five papers.

Technical validation is relatively well established, but evidence that AI changes real clinical outcomes remains limited. Shyr et al. explicitly identify the need for prospective evaluation of diagnostic accuracy, time-to-diagnosis, and clinician workload. Zhang et al. similarly describe the limited real-world adoption of AI systems.

Thus, the literature lacks sufficient **prospective, randomized, workflow-aware evaluations** measuring clinical effectiveness rather than merely model performance.

4.4 Practical and Applied Gaps

4.4.1 Disconnection between model output and clinical action

Most studies report a prediction, classification, segmentation, ranked phenotype, or explanation. Much less attention is given to how that output changes the clinician's next action.

A clinically useful system should answer not only:

> “What is the predicted condition?”

but also:

> “How certain is the prediction, what evidence supports it, what additional evidence is needed, and how should the result be incorporated into the clinical workflow?”

4.4.2 Limited human-in-the-loop evaluation

RARE-PHENIX explicitly identifies human-in-the-loop diagnosis as a potential application, but prospective evaluation of clinician interaction remains necessary.

Similarly, Sangwan recommends healthcare-professional user studies to determine whether explanations genuinely address clinical needs.

This indicates a substantial applied gap in **human-AI interaction design, clinician trust calibration, cognitive workload, and decision-support effectiveness**.

4.4.3 Lack of integrated privacy-preserving clinical infrastructure

Pati et al. demonstrate that federated learning can allow institutions to collaborate without centralizing patient data. However, federated learning is primarily evaluated as a training strategy rather than as one component of a complete clinical architecture integrating multimodal data, LLMs, vector retrieval, explainability, uncertainty estimation, model routing, and deployment governance.

4.4.4 Limited resource-aware deployment

Clinical deployment frequently occurs under computational and infrastructural constraints. Pati et al. explicitly consider low-resource inference and report runtime optimizations through quantization, while Shyr et al. identify very large LLMs as difficult to deploy in constrained environments.
However, systematic comparisons between cloud-based, institutional, edge, and hybrid deployment architectures remain limited.

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5. Consolidated Research-Gap Matrix

Gap Category	Evidence from Literature	Unresolved Problem	Research Opportunity
Methodological	Small datasets and inadequate external validation in earlier cancer ML studies	Performance may not generalize across institutions or populations	Multi-level external, temporal, geographic, and prospective validation
Methodological	Heterogeneous clinical, genomic, imaging, and text data	Modalities are often analyzed separately	Unified multimodal learning architecture
Methodological	Clinician-curated labels may be incomplete or variable	Ground truth uncertainty is under-modeled	Multi-expert annotation and uncertainty-aware supervision
Theoretical	Prediction, explainability, privacy, and workflow treated separately	No unified model of trustworthy clinical AI	Integrated trustworthy-AI framework
Theoretical	Pixel-level explanations may not map to clinical concepts	Computational explanation ≠ clinical explanation	Concept-based and clinically grounded explainability
Theoretical	Mostly static prediction models	Insufficient treatment of temporal change	Longitudinal and continual-learning framework
Empirical	Rare diseases and rare cancer subtypes remain data limited	Underrepresentation of uncommon conditions	Cross-disease and rare-subtype learning
Empirical	Geographic diversity does not necessarily establish demographic fairness	Unequal performance may remain hidden	Subgroup fairness and healthcare-system evaluation
Practical	Retrospective evaluation dominates	Clinical outcome benefit remains uncertain	Prospective workflow and outcome studies
Practical	AI outputs are often isolated predictions	Limited connection to subsequent clinical actions	Human-in-the-loop decision-support architecture

Practical	Large models can be computationally demanding	Deployment barriers in low-resource settings	Efficient, adaptive, resource-aware AI

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6. Overall Research Gap

The reviewed literature demonstrates that the principal limitation of contemporary clinical AI is no longer the absence of capable algorithms. Instead, the field lacks a **unified, clinically validated architecture capable of combining heterogeneous data while maintaining generalizability, privacy, interpretability, uncertainty awareness, computational efficiency, and workflow compatibility**.

The literature has progressively solved individual parts of this problem. Kourou et al. established the potential of ML for cancer prediction but emphasized sample-size and validation limitations. Pati et al. demonstrated that federated learning can enable large-scale geographically diverse training and stronger out-of-sample generalization. Sangwan introduced quantitative evaluation of explanation quality while recognizing that explanations must eventually become clinically meaningful. Shyr et al. demonstrated that modeling an end-to-end clinical workflow can outperform isolated phenotyping approaches while identifying the need for prospective clinical evaluation.

Consequently, the most significant opportunity is to move from **task-specific AI models** toward a **generalizable clinical intelligence framework** capable of dynamically selecting and integrating models according to patient data type, disease context, evidence availability, uncertainty, and clinical requirements.

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7. Proposed Research Agenda

Research Question 1: Multimodal Clinical Intelligence

RQ1: *Can a unified multimodal AI architecture integrating medical images, clinical notes, structured clinical variables, pathology, and molecular information improve diagnostic and prognostic performance compared with modality-specific models across multiple cancer types and rare cancer subtypes?*

Hypothesis H1: A multimodal architecture will demonstrate superior discrimination, calibration, robustness, and clinical utility compared with single-modality models, particularly when complementary information is distributed across different clinical data sources.

Research Question 2: Federated Multimodal Learning

RQ2: *Can privacy-preserving federated learning improve the cross-institutional generalizability of multimodal cancer AI models while maintaining clinically acceptable computational and communication costs?*

Hypothesis H2: Federated training across heterogeneous institutions will produce models with significantly stronger out-of-site generalization than single-institution training while preserving data locality and reducing the need for centralized patient-level data aggregation.

This research directly extends the multi-institutional success demonstrated by Pati et al. from imaging segmentation toward multimodal clinical intelligence.

Research Question 3: Clinically Grounded Explainability

RQ3: *Do concept-based, clinically grounded explanations improve clinician trust, diagnostic accuracy, and decision efficiency more effectively than conventional pixel- or feature-attribution explanations?*

Hypothesis H3: Explanations expressed in clinically meaningful concepts and linked to supporting evidence will produce higher clinician comprehension, appropriate reliance, and decision accuracy than conventional post-hoc attribution maps.

This directly addresses Sangwan's observation that pixel-level explanations may fail to correspond to clinically meaningful diagnostic reasoning.

Research Question 4: End-to-End Clinical Workflow

RQ4: *Can an end-to-end AI pipeline that performs input interpretation, evidence extraction, disease-specific model routing, knowledge retrieval, uncertainty assessment, explanation generation, and human review reduce diagnostic time and clinician workload without compromising diagnostic safety?*

Hypothesis H4: Workflow-oriented AI will provide greater clinical utility than isolated prediction models because its outputs are directly connected to the sequence of decisions clinicians must make.

This research builds conceptually on RARE-PHENIX, which demonstrated the value of integrating extraction, standardization, and prioritization rather than treating these tasks independently.

Research Question 5: Prospective Clinical Utility and Adaptive Learning

RQ5: *Does prospective deployment of an uncertainty-aware, continuously monitored clinical AI system improve diagnostic accuracy, time-to-diagnosis, treatment decision quality, and clinician workload across diverse healthcare environments?*

Hypothesis H5: Prospective, continuously monitored AI assistance will improve clinical outcomes relative to conventional workflows, provided that model uncertainty, subgroup performance, data drift, and clinician override behavior are explicitly monitored.

This question addresses the strongest remaining translational gap identified by the literature: retrospective performance does not establish real-world clinical benefit. Shyr et al. explicitly call for prospective studies examining diagnostic accuracy, time to diagnosis, gene/variant prioritization, and clinician workflow.

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8. Recommended Conceptual Direction for Future Research

The literature supports a future research architecture organized around six interacting layers:

Data Layer → Multimodal Learning Layer → Federated/Privacy Layer → Knowledge and Reasoning Layer → Explainability and Uncertainty Layer → Human Clinical Decision Layer

Such an architecture would address limitations identified across the five papers rather than solving only one isolated problem.

The **Data Layer** should support clinical notes, imaging, pathology, structured records, laboratory data, and molecular information.

The **Multimodal Learning Layer** should dynamically combine disease-specific prediction models rather than assuming one model is optimal for every input.

The **Federated/Privacy Layer** should enable institutions to collaboratively improve models without requiring centralized patient-level data, extending the principle demonstrated by Pati et al.

The **Knowledge and Reasoning Layer** should connect model outputs with structured biomedical knowledge, ontologies, guidelines, and evidence resources. RARE-PHENIX provides a useful precedent for ontology-grounded processing and retrieval-based standardization.

The **Explainability and Uncertainty Layer** should provide both quantitative explanation evaluation and clinically meaningful evidence presentation, addressing the concerns identified by Sangwan.

Finally, the **Human Clinical Decision Layer** should explicitly incorporate clinician review, disagreement, confidence, override behavior, and prospective outcome monitoring.

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9. Final Research Gap Statement

Despite substantial progress in machine learning, federated learning, large language models, and explainable AI for cancer and rare-disease applications, existing research remains fragmented across prediction, privacy, explainability, phenotyping, and workflow integration. There is insufficient evidence for a unified, multimodal, privacy-preserving, clinically interpretable,

uncertainty-aware, and prospectively validated AI framework capable of generalizing across institutions, populations, cancer types, and rare disease contexts while producing clinically actionable outputs. Future research should therefore shift from isolated model optimization toward end-to-end clinical AI systems evaluated not only by predictive accuracy but also by external generalizability, fairness, explanation quality, computational efficiency, workflow compatibility, and measurable clinical outcomes.

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10. Literature Base Used

Kourou, K., Exarchos, T. P., Exarchos, K. P., Karamouzis, M. V., & Fotiadis, D. I. (2015). *Machine learning applications in cancer prognosis and prediction.* Computational and Structural Biotechnology Journal, 13, 8–17.

Pati, S., et al. (2022). *Federated Learning Enables Big Data for Rare Cancer Boundary Detection.* The study evaluates federated learning using data from 71 institutions across six continents and 6,314 glioblastoma patients.

Zhang, B., Shi, H., & Wang, H. (2023). *Machine Learning and AI in Cancer Prognosis, Prediction, and Treatment Selection: A Critical Approach.* Journal of Multidisciplinary Healthcare, 16, 1779–1791.

Sangwan, H. (2024). *Quantifying Explainable AI Methods in Medical Diagnosis: A Study in Skin Cancer.* medRxiv preprint, posted December 10, 2024. The uploaded paper explicitly states that the manuscript had not undergone peer review at the time of posting.

Shyr, C., Hu, Y., Tinker, R. J., et al. (2026). *An artificial intelligence framework for end-to-end rare disease phenotyping from clinical notes using large language models.* The study presents and externally validates RARE-PHENIX across UDN and Vanderbilt data.